

Gibbs endpoint likelihood and exact production radial merge boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-105 left two possible continuations of the exact remaining $q = 10/9$ Nelson problem: an estimate along the actual Gibbs smart path, or a complete signed same-root cross-mode/forest packet. This note attacks both routes before introducing another owner decomposition.

The outcome is a boundary theorem with four parts.

1. The actual Gibbs path has an exact endpoint likelihood and relative- entropy representation. Its total time integral is precisely the unknown self-coupled versus decoupled free-energy difference. Thus thermodynamic integration alone is an identity, not an estimate.
2. At a constant active-doublet field the endpoint likelihood has a lower bound of order A^2N , where N is the physical sharp-cube Fourier radius. Taking $A = \sigma N^{1/4}$ defeats every fixed pointwise sextic plus Cameron–Martin payment. Entropy or full Gaussian integration cannot be discarded.
3. On an exact production radial fibre, a 1:2 same-root Fourier inclusion–exclusion has merge defect $-c\lambda^4 + O(\lambda^2)$. Hence a universal bounded inclusion–exclusion input-leaf correction of the form (6.11) is impossible. A leafwise sextic split is not superadditive either.
4. A surviving candidate square coordinate is the coherent *output*-frequency current obtained after the complete nonlinear production current is assembled. It must remain with its heat lift, covariance trace, mixed baseline, R-063 forest, and one parent/root sextic term.

The note also corrects two R-105 notation defects. The stabilising top-shell coefficient is $3L^3/64$, not the coefficient inherited from A9’s original physical potential, and the linearly divergent sharp-cube variable is the physical radius N , not a dyadic index J . Neither correction changes the R-105 ratio $3/t$, route verdicts, or tier.

All statements are finite-cutoff and fixed-positive-floor. The production merge theorem is deterministic and therefore rules out only a bounded pointwise input-leaf reconstruction. It is not a counterexample to the Gaussian/Gibbs integral, OVERLAP_{src}, Nelson, cutoff or floor removal, a measure theorem, or Sector-A closure. Tier stays T4.

2 Exact Gibbs path and endpoint likelihood

Fix a finite cutoff J and write $q = 10/9$. Throughout this note the R-093 stabilized potential and energy are

$$U_J(\phi) := \frac{3}{20} \|\phi\|_6^6, \quad G_J(x) := U_J(K_J x) + \frac{1}{2} \{x^T T_J(x)x - \text{Tr } T_J(x)\}. \quad (2.0)$$

Let $T_J(x) \succeq 0$ be the self-coupled coefficient operator and put

$$z_t = \sqrt{1-t}x + \sqrt{t}y, \quad 0 \leq t \leq 1, \quad (2.1)$$

for independent standard Gaussian sources x, y . The lifted Gibbs law is

$$d\hat{\nu}_{J,t}(x, y) = \frac{1}{Z_{J,t}} \exp \left[-qU_J(K_J x) - \frac{q}{2} \{z_t^T T_J(x) z_t - \text{Tr } T_J(x)\} \right] d\gamma_J(x) d\gamma_J(y). \quad (2.2)$$

Integrating y gives A9's marginal

$$d\nu_{J,t} = e^{W_{J,t} - \Phi_{J,t}} d\gamma_J, \quad Z_{J,t} = \mathbb{E}_{\gamma_J} e^{W_{J,t}}, \quad \Phi_{J,t} = \log Z_{J,t}. \quad (2.3)$$

The endpoints are

$$W_{J,0} = -qU_J + \frac{q}{2} \{\text{Tr } T_J - x^T T_J x\}, \quad (2.4)$$

$$W_{J,1} = -qU_J - \frac{1}{2} \log \det {}_2(\text{I} + qT_J) = -qU_J + \frac{q}{2} \text{Tr } T_J - \frac{1}{2} \log \det(\text{I} + qT_J). \quad (2.5)$$

Thus $t = 0$ is the self-coupled endpoint and $t = 1$ is the controlled A8 independent-carrier endpoint.

Theorem 2.1 (exact endpoint likelihood and entropy identities)

Define

$$\mathcal{L}_J(x) := W_{J,0}(x) - W_{J,1}(x) = \frac{1}{2} \log \det(\text{I} + qT_J(x)) - \frac{q}{2} x^T T_J(x) x. \quad (2.6)$$

The potential and the full linear trace cancel exactly. Moreover

$$\frac{d\nu_{J,0}}{d\nu_{J,1}} = \exp\{\mathcal{L}_J - (\Phi_{J,0} - \Phi_{J,1})\}, \quad (2.7)$$

and therefore

$$\Phi_{J,0} - \Phi_{J,1} = \mathbb{E}_{\nu_{J,0}} \mathcal{L}_J - H(\nu_{J,0} \mid \nu_{J,1}) \quad (2.8)$$

as well as

$$\Phi_{J,0} - \Phi_{J,1} = \mathbb{E}_{\nu_{J,1}} \mathcal{L}_J + H(\nu_{J,1} \mid \nu_{J,0}). \quad (2.9)$$

More generally the Gibbs variational formula gives

$$\sup_{\nu} \{\mathbb{E}_{\nu} \mathcal{L}_J - H(\nu \mid \nu_{J,1})\} = \log \mathbb{E}_{\nu_{J,1}} e^{\mathcal{L}_J} = \Phi_{J,0} - \Phi_{J,1}, \quad (2.10)$$

and the optimiser is $\nu_{J,0}$.

Proof. Subtract (2.5) from (2.4) to obtain (2.6). Dividing the two normalized densities in (2.3) gives (2.7). Integrating its logarithm first against $\nu_{J,0}$ and then against $\nu_{J,1}$ proves (2.8)–(2.9). Exponentiating (2.7) gives $Z_{J,0}/Z_{J,1} = \mathbb{E}_{\nu_{J,1}} e^{\mathcal{L}_J}$, and the Gibbs variational formula proves (2.10). $\square$

The signs in (2.8)–(2.9) are load-bearing. Forward entropy has a minus sign but is taken under the unknown self-coupled Gibbs law. Reverse entropy is available from the controlled endpoint only with a plus sign and hence does not upper-bound the desired free energy.

3 Why total time integration is not a new estimate

Let $\mathfrak{B}_{J,t}$ denote the complete A9 bracket from A9-(3.8). At every finite cutoff ordinary differentiation gives

$$\Phi'_{J,t} = \frac{q}{2} \mathbb{E}_{\nu_{J,t}} \mathfrak{B}_{J,t}. \quad (3.1)$$

Consequently

$$\frac{q}{2} \int_0^1 \mathbb{E}_{\nu_{J,t}} \mathfrak{B}_{J,t} dt = \Phi_{J,1} - \Phi_{J,0}. \quad (3.2)$$

Equation (3.2) is exact thermodynamic integration. It does not supply a lower bound for its right side.

The independent endpoint is bounded on both sides uniformly in the cutoff. A8 gives the upper bound. Since $T_J \succeq 0$ implies $-\frac{1}{2} \log \det_2(\mathbf{I} + qT_J) \geq 0$, Jensen gives

$$Z_{J,1} \geq \mathbb{E} e^{-qU_J} \geq e^{-q\mathbb{E}U_J}, \quad (3.3)$$

and the regulated point covariance makes $\sup_J \mathbb{E}U_J < \infty$. Therefore a uniform lower bound for the integral in (3.2) is equivalent, up to this already controlled endpoint, to the missing upper bound for $\Phi_{J,0}$.

The same boundary appears in R-093 source coordinates. There $\nu_{J,0} = \nu_J^*$ and, for a temporally faithful source chart h with fibre entropy Φ_h ,

$$q\mathcal{A}_J(h) = -\Phi_{J,1} + H(\nu_h \mid \nu_{J,1}) - \mathbb{E}_{\nu_h} \mathcal{L}_J + \Phi_h. \quad (3.4)$$

Taking the directed-union infimum of the last three terms gives exactly $\Phi_{J,1} - \Phi_{J,0}$. R-093 near minimisers make the excess $H(\nu_h \mid \nu_{J,0}) + \Phi_h$ tend to zero. This is the only nonnegative excess over $-\Phi_{J,0}$. The term $H(\nu_h \mid \nu_{J,1})$ is not an independent reserve because (3.4) pairs it exactly with $-\mathbb{E}_{\nu_h} \mathcal{L}_J$.

This fires the narrow route verdict **NG-2026-07-28-A13-TOTAL-A9-TIME-INTEGRATION-IDENTITY**: reporting (3.2), (2.10), or (3.4) as an estimate without a separately proved root-local production lower bound is circular. A Gibbs-only *local* differential estimate or a root-summed signed estimate remains open.

4 Pointwise endpoint-likelihood coercivity fails

Let $Q_N = \{n \in \mathbb{Z}^3 : |n|_\infty \leq N\}$ be an admitted common sharp-cube cutoff. The letter N here is the physical Fourier radius; in a dyadic convention $N = 2^J$. Let u be a real active-doublet unit vector and choose the constant physical configuration

$$K_N x_A = Au. \quad (4.1)$$

Then $DK_N x_A = 0$, hence

$$x_A^T T_N(x_A) x_A = 0. \quad (4.2)$$

The accepted two-sided production covariance ellipticity gives, on every outer-shell mode $N/2 \leq |n|_\infty \leq N$,

$$\widehat{C}_N(k_n) \succeq c_C \langle k_n \rangle^{-4} \mathbf{I}. \quad (4.3)$$

The exact production radial formula proved again in Section 5 gives an active eigenvalue at least $4aA^2$. Since T_N is Fourier block diagonal at the constant field, every outer block has an eigenvalue at least $c_1 A^2/N^2$. There are at least $c_{\text{sh}} N^3$ retained real degrees of freedom for a fixed $c_{\text{sh}} > 0$, and for A^2/N^2 sufficiently small, $\log(1+s) \geq s/2$ gives

$$\log \det(\mathbf{I} + qT_N(x_A)) \geq c_2 A^2 N. \quad (4.4)$$

Equations (2.6), (4.2), and (4.4) imply

$$\mathcal{L}_N(x_A) \geq c_3 A^2 N. \quad (4.5)$$

The configuration x_A uses only the zero mode. Since the admitted covariance has $\widehat{C}(0) > 0$, its Cameron–Martin norm obeys $\|x_A\|^2 = c_{\text{CM}} A^2$ with c_{CM} independent of N . Take $A = \sigma N^{1/4}$. For any fixed $\zeta, \eta \geq 0$,

$$\mathcal{L}_N(x_A) - \zeta \|Au\|_6^6 - \eta \|x_A\|^2 \geq N^{3/2} \{c_3 \sigma^2 - \zeta L^3 \sigma^6\} - O_\eta(N^{1/2}). \quad (4.6)$$

Choosing $\sigma > 0$ sufficiently small makes (4.6) tend to $+\infty$. Therefore no cutoff-uniform pointwise upper bound of the likelihood by any fixed sextic plus Cameron–Martin payment exists.

This fires **NG-2026-07-28-A13-POINTWISE-ENDPOINT-LIKELIHOOD-COERCIVITY**. The constant field is a Gaussian-null configuration. Suppressing all nonconstant modes has an entropic cost invisible in a pointwise inequality, so (4.6) is not a Gibbs or Nelson counterexample.

5 Exact production radial coefficient

Use the pinned fixed floor $\epsilon = 10^{-12}$ and

$$P = 4 + 10^{-12}, \quad a = \frac{9}{500P}, \quad b = \frac{3}{400P}, \quad c = \frac{3}{320P}, \quad d = a + 2b + c. \quad (5.1)$$

Let E_Z be the active four-real doublet projector, let J be its phase rotation, and fix a real active-doublet unit vector u . On $X = fu$, put $y = f^2$ and $R = y + \epsilon$. The accepted A13 Fierz formula gives

$$B(fu) = yB_\infty(u) + \delta_\epsilon(f)u \otimes u, \quad (5.2)$$

$$B_\infty(u) = 4d(E_Z - Ju \otimes Ju) - (8b + 4c)u \otimes u, \quad (5.3)$$

$$\delta_\epsilon(f) = \frac{8b\epsilon y}{y + \epsilon} + \frac{4c\epsilon^2 y}{(y + \epsilon)^2}. \quad (5.4)$$

In particular

$$B_\infty(u)u = 4au, \quad 0 \leq \delta_\epsilon(f) \leq K_\epsilon, \quad K_\epsilon := \epsilon(8b + c). \quad (5.5)$$

Proof. The radial Fierz identities are

$$\mathcal{P} = 4y\{E_Z - (Ju) \otimes (Ju)\}, \quad Z \otimes X + X \otimes Z = 2yu \otimes u, \quad X \otimes X = yu \otimes u. \quad (5.6)$$

Substitution and collection of the zero-floor part gives (5.2)–(5.4). Since $4d - (8b + 4c) = 4a$, (5.5)'s eigenvalue follows. Finally $y/(y + \epsilon) \leq 1$ and $4\epsilon y \leq (y + \epsilon)^2$ prove the upper bound. $\square$

There is an independent check from the R-082 global square. With

$$c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}, \quad \alpha = \frac{5}{9}, \quad (5.7)$$

its radial eigenvalue is

$$4y \left[c_0 + c_1 \left(1 - \alpha \frac{y}{y + \epsilon} \right)^2 \right], \quad (5.8)$$

and the exact identity

$$c_0 + c_1(1 - \alpha t)^2 = a + 2b(1 - t) + c(1 - t)^2 \quad (5.9)$$

reproduces (5.2).

6 The production 1:2 merge obstruction

Let k and $2k$ both lie inside one fixed cutoff, put $\theta = k \cdot x$, and use normalized torus average $\langle \cdot \rangle$. For fixed positive-semidefinite derivative covariance matrices Γ_i , define

$$V_\Gamma(X) = \frac{1}{2} \left\langle \sum_i (\partial_i X)^T B(X) \partial_i X - \sum_i B(X) : \Gamma_i \right\rangle, \quad (6.1)$$

$$g_{\Gamma,u} := \sum_i u^T \Gamma_i u < \infty. \quad (6.2)$$

For fixed $r \neq 0$ set

$$F = 1 + r \cos \theta - \cos 2\theta, \quad F_1 = 1 + r \cos \theta, \quad F_2 = 1 - \cos 2\theta, \quad F_0 = 1. \quad (6.3)$$

The inclusion–exclusion merge defect is

$$\Delta_\lambda := V_\Gamma(\lambda F u) - V_\Gamma(\lambda F_1 u) - V_\Gamma(\lambda F_2 u) + V_\Gamma(\lambda F_0 u). \quad (6.4)$$

Theorem 6.1 (exact production radial merge boundary)

For every $\lambda > 0$,

$$\Delta_\lambda \leq -\frac{a|k|^2 r^2}{2} \lambda^4 + \frac{K_\epsilon |k|^2 (r^2 + 4)}{2} \lambda^2 + K_\epsilon g_{\Gamma, u}. \quad (6.5)$$

In particular $\Delta_\lambda \rightarrow -\infty$ as $\lambda \rightarrow \infty$. More explicitly, if

$$A = \frac{a|k|^2 r^2}{2}, \quad B = \frac{K_\epsilon |k|^2 (r^2 + 4)}{2}, \quad C = K_\epsilon g_{\Gamma, u}, \quad (6.6)$$

then $\Delta_\lambda < -M$ whenever

$$\lambda^2 > \frac{B + \sqrt{B^2 + 4A(C + M)}}{2A}. \quad (6.7)$$

Proof. Exact trigonometric integration gives

$$\langle F^2 |F'|^2 - F_1^2 |F'_1|^2 - F_2^2 |F'_2|^2 \rangle = -\frac{r^2}{4}, \quad (6.8)$$

$$\langle F^2 - F_1^2 - F_2^2 + F_0^2 \rangle = 0, \quad (6.9)$$

and

$$\langle |F'|^2 + |F'_1|^2 + |F'_2|^2 \rangle = r^2 + 4. \quad (6.10)$$

By (5.5), the zero-floor raw defect is exactly $-(a|k|^2 r^2/2)\lambda^4$, including the physical outer factor $1/2$. Its covariance trace cancels by (6.9). Bounding every rational raw term in absolute value by K_ϵ and using (6.10) gives the quadratic term in (6.5). Four numbers in $[0, K_\epsilon]$ have the signed inclusion–exclusion magnitude at most $2K_\epsilon$; the outer $1/2$ leaves the last term in (6.5). Solving the quadratic inequality in λ^2 proves (6.7). $\square$

Thus there is no finite fixed-cutoff constant K_J for which

$$V_\Gamma(\lambda F u) \geq V_\Gamma(\lambda F_1 u) + V_\Gamma(\lambda F_2 u) - V_\Gamma(\lambda F_0 u) - K_J \quad (6.11)$$

holds at every amplitude. This fires NG-2026-07-28-A13-PRODUCTION-INPUT-MODE-MERGE-TENSORIZATION. It strengthens R-105's scalar sign diagnostic to the exact production coefficient, same-root covariance-normal coordinate.

The stabilising sextic cannot be split among these leaves either. Exact integration gives

$$\langle F^6 - F_1^6 - F_2^6 + F_0^6 \rangle = -\frac{15r^2(9r^2 + 2)}{32} < 0. \quad (6.12)$$

This does not say that the physical sextic or complete action is negative. It says only that a leafwise inclusion–exclusion duplicates the sextic with the wrong superadditivity sign. Keeping the sextic once at the parent/root remains viable and is mandatory for this candidate route.

7 A coherent output-frequency square is a surviving coordinate

R-082 Theorem 7.1 gives the exact global Pauli–Fierz square

$$\frac{1}{2} \sum_i \|\Xi(X, \partial_i X)\|_{L^2}^2 = \frac{1}{2} \sum_{i,n} |\Xi(\widehat{X}, \partial_i X)(n)|^2. \quad (7.1)$$

This is genuine nonnegative additivity for disjoint *output*-frequency projectors, applied only after the full nonlinear current is formed. It is not additivity over input modes or Gaussian roots. Different input

convolutions can land at the same output n and interfere before the square; Theorem 6.1 is an exact such interference fixture.

The covariance trace in (6.1) is not itself an output square. Moreover the coefficient $C(X)$ in $B(X) = C(X)^T C(X)$ is same-root and state dependent. It therefore lies outside the determinant theorem for a predictable coefficient and a fresh derivative. The valid root-local successor must preserve, for every output n ,

$$\hat{\Xi}(n) = \sum_{\alpha \rightarrow n} \hat{\Xi}_\alpha(n), \quad (7.2)$$

where α ranges over every input convolution reaching n , before squaring. The proof must retain the complete heat lift, covariance trace, mixed baseline, R-063 forest, and parent/root sextic in the same signed packet.

No current authority bounds this residual. R-077 cancels a complete fresh packet before projection; R-096 compresses a same-root coefficient to Hermite ranks 0, 1, 2 but supplies no spatial/root gain; R-097 retains its posterior covariance deficit; and A9-(3.8) retains the same-root Gibbs bracket. R-089/R-099 control the Cameron–Martin coordinates, not this nonlinear coefficient.

The narrowest currently identified analytic target in the audited factorisation coordinate is a root-local, actual-Gibbs time-integrated bound for the complete coherent-output packet (7.2), with a summable root loss. The equivalent direct complete signed source-action route remains open; this section does not assert uniqueness. In either formulation, do not split the packet into input-mode leaves, do not split the sextic, and do not count the total endpoint integral (3.2) as the estimate.

8 R-105 correction audit

R-105 Proposition 6.2 used the stabilized Nelson functional

$$G_J = V_J^{\text{ren}} + \frac{3}{20} \|\phi\|_6^6. \quad (8.1)$$

For the top-shell wave $g = \cos(k \cdot x)$, $\langle g^6 \rangle = 5/16$, so R-105-(6.20) must read

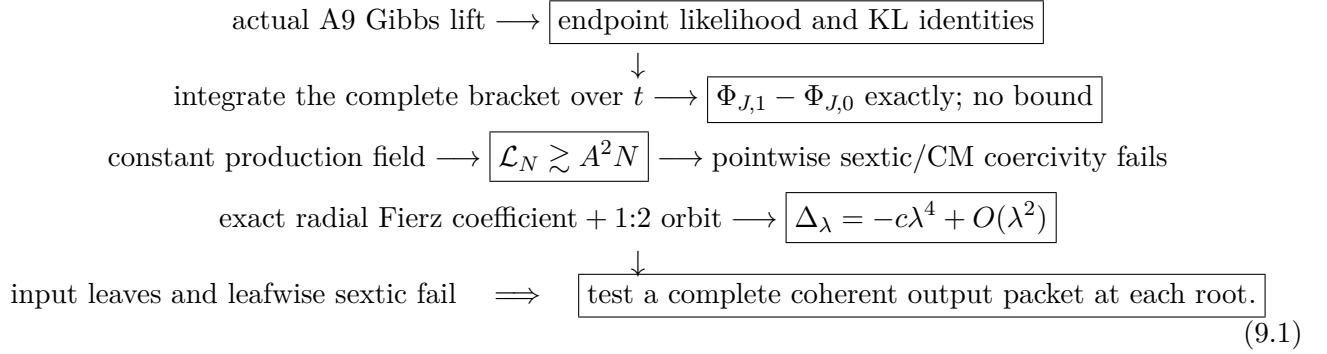
$$U_6(Ae) = u_6 A^6, \quad \boxed{u_6 = \frac{3}{20} \frac{5}{16} L^3 = \frac{3L^3}{64} > 0.} \quad (8.2)$$

The former $5\gamma L^3/96$ belonged to A9's original $(\gamma/6)\|\phi\|_6^6$ physical potential and was imported into the wrong functional. Euler homogeneity still gives $\langle DU_6(e), e \rangle = 6u_6$, and both the bracket and free-energy leading terms contain the same u_6 . Hence the forced ratio $b(t) \geq 3/t$ and every R-105 verdict are unchanged.

R-105-(7.1)–(7.3) also used the letter J for the physical sharp-cube radius. The authority is linear in N for Q_N , with $N = 2^J$ under dyadic indexing. The corrected proof note uses N and leaves the stored slope and method no-go unchanged.

This audit is registered as AUDIT-2026-07-28-A13-R105-SEXTIC-COEFFICIENT-CUTOFF-NOTATION.

9 Attempt, failure, and evidence map



Route	Verdict	Exact reason / successor
Total A9 time integral	failed as a proof	It is the endpoint free-energy difference (3.2); prove a root-local lower bound first.
Endpoint KL reserve	boundary only	Forward entropy is under the unknown self Gibbs law; reverse entropy has the wrong sign.
Pointwise likelihood payment	failed	The constant ray has $\mathcal{L}_N \gtrsim A^2 N$ and defeats fixed sextic/CM payment.
Bounded inclusion–exclusion input-leaf reassembly	failed	Exact production merge defect tends to $-\infty$ by Theorem 6.1.
Leafwise sextic repair	failed	The exact merge (6.12) is negative.
Coherent output packet	open candidate	Form the complete current first; retain heat, trace, forest, baseline, and sextic once at the root.

10 Devil’s-advocate review

1. **DISMISSED** – “The endpoint KL identity proves Nelson because entropy is nonnegative.” In (2.8) entropy is subtracted but the expectation is under the unknown self-coupled law. In (2.9) the controlled law comes with positive entropy. The variational identity (2.10) is exactly the missing free energy.
2. **DISMISSED** – “The all-law top-shell no-go already refutes the actual Gibbs path.” R-105 used narrow translated Gaussian laws. The actual $\nu_{J,t}$ is the marginal of (2.2) and retains the same-root x – z_t correlations. Only an unconditional all-law pointwise theorem was refuted.
3. **VALID WITH MITIGATION** – “The constant ray is Gaussian-null.” Correct. Section 4 is a method no-go for pointwise extraction only. The candidate proof must keep entropy or perform the full root integration.
4. **VALID WITH MITIGATION** – “A negative merge defect is not a negative complete action.” Correct. Every individual raw production square remains nonnegative and has positive quartic growth. Theorem 6.1 rejects only bounded input-leaf reassembly.
5. **DISMISSED** – “The positive sextic repairs the input-leaf factorisation automatically.” Equation (6.12) has the opposite inclusion–exclusion sign. The sextic remains useful only when kept once at the parent/root.

6. **UPHELD** – “No cutoff-uniform coherent-output Gibbs estimate is proved.” This is the exact remaining obstruction. R-106 narrows its legal coordinate but does not bound it; Nelson and Sector A remain open.

11 Executable verification

The primary certificate derives the endpoint algebra, corrected sextic coefficient, physical-radius scaling, exact production radial eigenvalue, R-082 radial identity, all trigonometric merges, the divergence threshold, and the coherent-output cross terms symbolically. A non-importing standard-library implementation checks the endpoint KL identities on an unrelated finite law and computes the Fourier averages with exact rational Laurent polynomials.

Run

```
E:\Dev\TECT.venv\Scripts\python.exe '
codes/foundations/a13_classii_gibbs_endpoint_production_merge_boundary_verify.py
```

from the repository root. The integrated verifier reruns both implementations, checks the note, PDF, manifest, authority pins, public surfaces, exploration records, negative registry, and the R-105 correction.

Result footer

```
Result ID: A13-CLASSII-GIBBS-ENDPOINT-PRODUCTION-MERGE-BOUNDARY
Claim:      A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION
Ledger ID: R-106
Status:     T4 ANALYTIC/EXACT/EXECUTED BOUNDARY; NELSON AND SECTOR A OPEN
Scope:      Fixed cutoff/floor; q=10/9 Gibbs endpoints; active-doublet radial
            fibre; admitted k, 2k; fixed PSD merge covariance.
Precise statement: exact endpoint likelihood/KL/variation and time-integral
            boundary; pointwise likelihood no-go; exact radial coefficient;
            unbounded 1:2 input merge and leafwise-sextic non-superadditivity.
Identified (not bounded): a complete coherent output-frequency square with
            heat, trace, forest, baseline, and one parent/root sextic.
Dependencies: A9; R-063, R-077, R-082, R-089, R-093, R-096, R-097,
            R-099, R-104, R-105.
Evidence grade: ANALYTIC/EXACT/EXECUTED; TWO NON-IMPORTING ROUTES.
Reproduction command:
            E:\Dev\TECT.venv\Scripts\python.exe '
            codes/foundations/a13_classii_gibbs_endpoint_production_merge_boundary_verify.py
Expected output:
            Primary 46/46; independent 59/59; integrated 337/337;
            aggregate 442/442 PASS.
Negative results: NG-2026-07-28-A13-TOTAL-A9-TIME-INTEGRATION-IDENTITY;
            NG-2026-07-28-A13-POINTWISE-ENDPOINT-LIKELIHOOD-COERCIVITY;
            NG-2026-07-28-A13-PRODUCTION-INPUT-MODE-MERGE-TENSORIZATION.
Correction audit: R-105 reissued with  $u_6=3 L^{3/64}$  and physical radius N;
            ratio 3/t and route verdicts unchanged.
Falsification gate: failure of endpoint/KL/time orientation, ray scaling,
            radial Fierz or merge identities, any evidence/hash/PDF/ledger/
            public-surface contract, or any scope firewall.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement:
            no root-local Gibbs or coherent-output lower bound is proved;
            the direct source-action route is not excluded; no OVERLAP_src,
            Nelson, removal, measure, Sector-A, or T5--T7 conclusion follows.
Next required action: prove one summable root-local coherent-output estimate
            or the direct source-action bound, retaining one root sextic.
```